

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

Code of Conduct

I K. Sampath Reddy (192424285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sampath Reddy

1. Detection Performance Comparison

Given: Model A (Viola-Jones) — TP=80, FP=20, FN=30. Model B (YOLO) — TP=90, FP=15, FN=20.

$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN)$$

(a) Computation:

Model	Precision	Recall
A (Viola-Jones)	$80 / (80+20) = 0.80$ (80%)	$80 / (80+30) = 0.727$ (72.7%)
B (YOLO)	$90 / (90+15) = 0.857$ (85.7%)	$90 / (90+20) = 0.818$ (81.8%)

(b) Comparison and Conclusion:

Comparison: Model B outperforms Model A on both metrics — precision is higher by 5.7 points (fewer false positives relative to detections) and recall is higher by 9.1 points (fewer missed detections).

Evaluation: Since B dominates A on both precision and recall simultaneously (not a trade-off case), there is no ambiguity in ranking — B is unambiguously the stronger detector on this data.

Justification: Model B (YOLO) is better for object detection here because it achieves both higher precision and higher recall than Model A, meaning it produces fewer false alarms and misses fewer true objects at the same time.

4. Depth Estimation Comparison (Stereo vs SfM)

Given: Actual depths A=2 m, B=3 m, C=4 m. Stereo Vision: A=2.0, B=3.0, C=4.0. Structure from Motion (SfM): A=2.5, B=3.8, C=5.0.

$Absolute\ Error = |Estimated - Actual|$

(a) Computation:

Object	Actual	Stereo Estimate	Stereo Abs. Error	SfM Estimate	SfM Abs. Error
A	2.0	2.0	0.0	2.5	0.5
B	3.0	3.0	0.0	3.8	0.8
C	4.0	4.0	0.0	5.0	1.0

Average absolute error — Stereo Vision: $(0.0+0.0+0.0)/3 = 0.0$ m. Structure from Motion: $(0.5+0.8+1.0)/3 = 0.767$ m.

(b) Comparison and Conclusion:

Comparison: Stereo Vision matches the actual depth exactly for every object (zero error), while SfM consistently over-estimates depth, with error growing as actual depth increases (0.5 m → 0.8 m → 1.0 m).

Evaluation: SfM's increasing error trend suggests a scale-drift pattern common in monocular reconstruction, whereas Stereo Vision's dual-camera triangulation gives it a clear, consistent accuracy advantage in this dataset.

Justification: Stereo Vision is more accurate here, with a 0.767 m lower average absolute error than SfM; it should be preferred whenever a second camera is available, while SfM remains a fallback only when hardware restricts the system to a single camera (as in Q5 of AT2).

5. Pattern Recognition Accuracy Comparison

Given: Traditional — 85 correct out of 100. Deep Learning — 92 correct out of 100.

Accuracy = Correct Predictions / Total Samples

(a) Computation:

System	Accuracy
Traditional	85 / 100 = 0.85 (85%)
Deep Learning	92 / 100 = 0.92 (92%)

(b) Comparison and Conclusion:

Comparison: Deep Learning achieves 7 percentage points higher accuracy than the Traditional system on the same sample size (100), making the comparison directly proportional and free of sample-size bias.

Evaluation: A 7-point accuracy gap on identical sample sizes is a meaningful, non-marginal difference, indicating Deep Learning generalizes pattern recognition more effectively on this dataset.

Justification: The Deep Learning system performs better and is the preferable choice for pattern recognition accuracy; the Traditional system may still be considered only if speed or computational cost outweighs the accuracy loss in the target deployment.

7. F1-Score Comparison

Given: Model A — Precision=0.80, Recall=0.60. Model B — Precision=0.75, Recall=0.75.

$$F1\text{-Score} = 2 \times (Precision \times Recall) / (Precision + Recall)$$

(a) Computation:

Model	Calculation	F1-Score
A	$2 \times (0.80 \times 0.60) / (0.80 + 0.60)$ $= 0.96 / 1.40$	0.686 (68.6%)
B	$2 \times (0.75 \times 0.75) / (0.75 + 0.75)$ $= 1.125 / 1.50$	0.750 (75.0%)

(b) Comparison and Conclusion:

Comparison: Model A has higher precision (0.80 vs 0.75) but noticeably lower recall (0.60 vs 0.75), while Model B has equal precision and recall. Despite A's precision edge, its F1-score (0.686) is lower than B's (0.750) because F1 penalizes the imbalance between precision and recall more than it rewards a high value in only one of them.

Evaluation: F1-score is the harmonic mean specifically because it punishes large gaps between precision and recall; Model A's 0.20 gap drags its F1 down despite a strong precision figure, while Model B's balanced 0.75/0.75 split maximizes F1 for that combined precision-recall total.

Justification: Model B is better overall because its higher, balanced F1-score (0.750 vs 0.686) reflects a more even trade-off between avoiding false positives and avoiding false negatives, which is generally more desirable than Model A's skew toward precision at the cost of missed detections.

12. IOU Comparison

Given IOU (Intersection over Union) values for three detected objects:

Object	Method A (IOU)	Method B (IOU)
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

$$\text{Average IOU} = (IOU1 + IOU2 + IOU3) / 3$$

(a) Computation:

Method A: $(0.60 + 0.70 + 0.65) / 3 = 1.95 / 3 = 0.65$

Method B: $(0.80 + 0.85 + 0.90) / 3 = 2.55 / 3 = 0.85$

(b) Comparison and Conclusion:

Comparison: Method B outperforms Method A on every individual object (0.80 vs 0.60, 0.85 vs 0.70, 0.90 vs 0.65) and its average IOU (0.85) is 0.20 higher than Method A's (0.65), a consistent and substantial margin.

Evaluation: Since IOU measures how tightly a predicted bounding box overlaps the ground-truth box, Method B's consistently higher values (all above the common 0.8 'good localization' threshold) indicate noticeably more precise bounding-box localization than Method A, whose values hover around 0.6–0.7.

Justification: Method B provides higher-quality detections and should be preferred, since its bounding boxes align much more closely with ground truth across all three objects, directly translating to better localization quality in any downstream detection pipeline.